

Lab Assignment # 06

Course Title : **AI Assistant Coding**
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Batch No. : **48**

Lab 6: AI-Based Code Completion – Classes, Loops, and Conditionals

Task Description #1: Classes (Student Class)

Scenario: You are developing a simple student information management module.

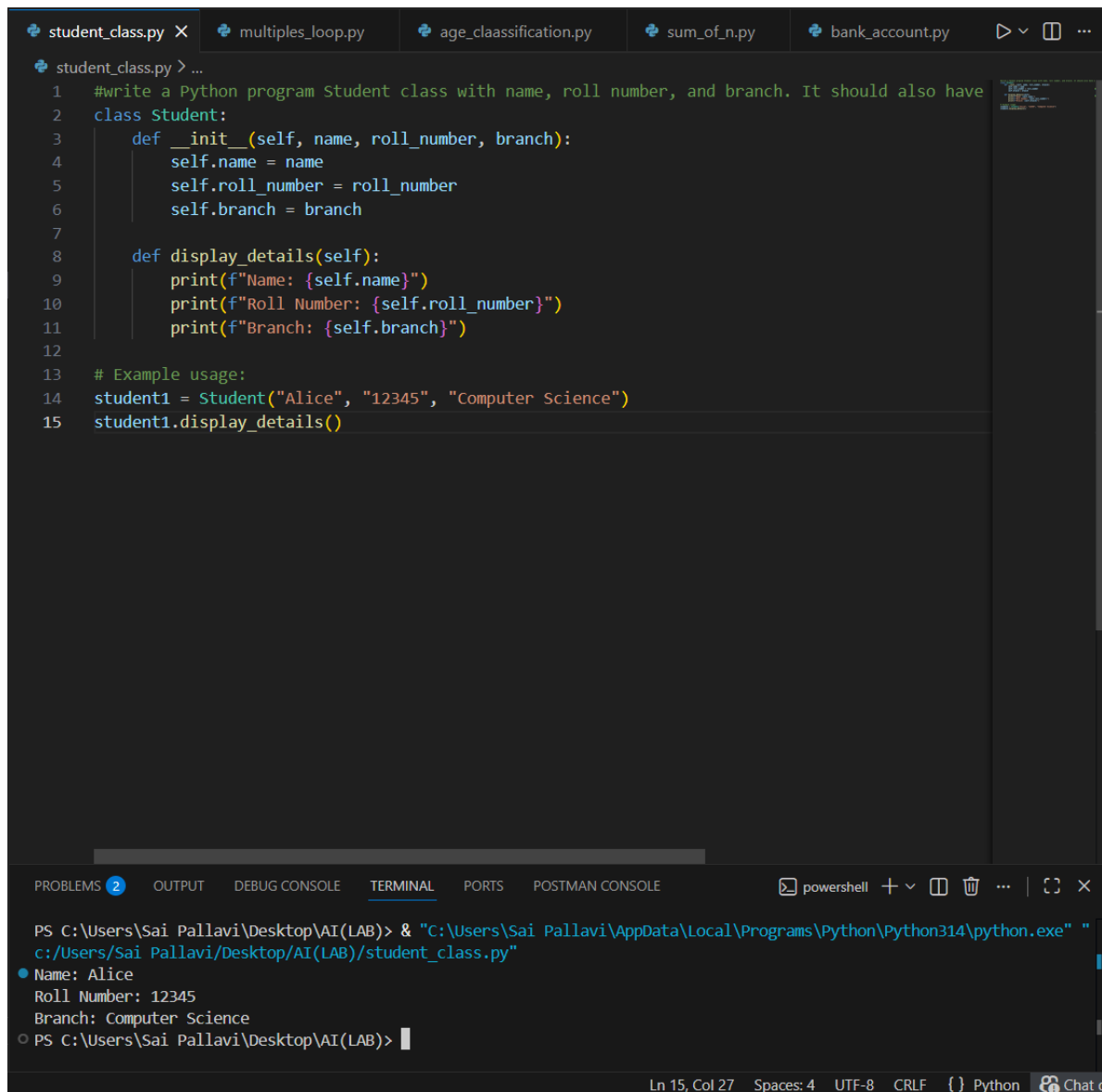
Task:

- Use an AI tool (GitHub Copilot / Cursor AI / Gemini) to complete a Student class.
- The class should include attributes such as name, roll number, and branch.
- Add a method `display_details()` to print student information.
- Execute the code and verify the output.
- Analyze the code generated by the AI tool for correctness and clarity.

Expected Output #1

- A Python class with a constructor (`__init__`) and a `display_details()` method.
- Sample object creation and output displayed on the console.
- Brief analysis of AI-generated code.

Output Screenshot:



The screenshot shows a Python IDE with several tabs: student_class.py, multiples_loop.py, age_classification.py, sum_of_n.py, and bank_account.py. The active tab is student_class.py, which contains the following code:

```
1 #write a Python program Student class with name, roll number, and branch. It should also have
2 class Student:
3     def __init__(self, name, roll_number, branch):
4         self.name = name
5         self.roll_number = roll_number
6         self.branch = branch
7
8     def display_details(self):
9         print(f"Name: {self.name}")
10        print(f"Roll Number: {self.roll_number}")
11        print(f"Branch: {self.branch}")
12
13 # Example usage:
14 student1 = Student("Alice", "12345", "Computer Science")
15 student1.display_details()
```

The terminal output shows the execution of the program:

```
PS C:\Users\Sai Pallavi\Desktop\AI(LAB)> & "C:\Users\Sai Pallavi\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Sai Pallavi/Desktop/AI(LAB)/student_class.py"
• Name: Alice
  Roll Number: 12345
  Branch: Computer Science
○ PS C:\Users\Sai Pallavi\Desktop\AI(LAB)>
```

Task Description #2: Loops (Multiples of a Number)

Scenario: You are writing a utility function to display multiples of a given number.

Task:

- Prompt the AI tool to generate a function that prints the first 10 multiples of a given number using a loop.
- Analyze the generated loop logic.
- Ask the AI to generate the same functionality using another controlled looping structure (e.g., while instead of for).

Expected Output:

- Correct loop-based Python implementation.
- Output showing the first 10 multiples of a number.
- Comparison and analysis of different looping approaches.

Output Screenshot:

```
student_class.py  multiples_loop.py X  age_classification.py  sum_of_n.py  bank_account.py  ▶ ▢ ...
multiples_loop.py > ...
2 def print_multiples(number):
6     Parameters:
7     number (int): The number for which to print the multiples.
8     """
9     for i in range(1, 11): # Loop from 1 to 10
10         multiple = number * i # Calculate the multiple
11         print(f"{number} x {i} = {multiple}") # Print the multiple in a formatted way
12 # Example usage:
13 if __name__ == "__main__":
14     num = 5 # You can change this number to print multiples of a different number
15     print(f"First 10 multiples of {num}:")
16     print_multiples(num)
17 # Example usage:
18 if __name__ == "__main__":
19     num = 5 # You can change this number to print multiples of a different number
20     print(f"First 10 multiples of {num}:")
21     print_multiples(num)
22 # Example usage:
23 if __name__ == "__main__":
24     num = 5 # You can change this number to print multiples of a different number
25     print(f"First 10 multiples of {num}:")
26     print_multiples(num)
27 # Example usage:
28 if __name__ == "__main__":
29     num = 5 # You can change this number to print multiples of a different number
30     print(f"First 10 multiples of {num}:")
31     print_multiples(num)
32
33
```

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```
PS C:\Users\Sai Pallavi\Desktop\AI(LAB)> & "C:\Users\Sai Pallavi\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Sai Pallavi/Desktop/AI(LAB)/multiples_loop.py"
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
PS C:\Users\Sai Pallavi\Desktop\AI(LAB)>
```

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Analysis of for Loop Logic:

- The for loop uses range(1, 11) to ensure the loop runs exactly 10 times.
- In each iteration, the loop variable i is multiplied with the given number n to produce the next multiple.
- The logic is simple and easy to follow, which improves code readability.
- This approach is ideal when the number of iterations is fixed and known beforehand.
-

Analysis of while Loop Logic:

- The while loop keeps running as long as the condition i <= 10 remains true
- The counter variable i must be initialized before the loop and updated manually inside the loop.
- This loop provides greater control over how and when the loop executes.
- It can be slightly error-prone if the condition or increment statement is not handled correctly, which may lead to infinite loops.

Task Description #3: Conditional Statements (Age Classification)

Scenario: You are building a basic classification system based on age.

Task:

- Ask the AI tool to generate nested if-elif-else conditional statements to classify age groups (e.g., child, teenager, adult, senior).
- Analyze the generated conditions and logic.
- Ask the AI to generate the same classification using alternative conditional structures (e.g., simplified conditions or dictionary-based logic).

Expected Output:

- A Python function that classifies age into appropriate groups.
- Clear and correct conditional logic.
- Explanation of how the conditions work

Output Screenshot:

The screenshot shows a Python IDE with several files open: 'student_class.py', 'multiples_loop.py', 'age_classification.py' (selected), 'sum_of_n.py', and 'bank_account.py'. The 'age_classification.py' file contains the following code:

```

38     age_group = classify_age(age)
39     print(f"You are classified as: {age_group}")
40 # Example usage:
41 if __name__ == "__main__":
42     age = int(input("Enter your age: "))
43     age_group = classify_age(age)
44     print(f"You are classified as: {age_group}")
45 # Example usage:
46 if __name__ == "__main__":
47     age = int(input("Enter your age: "))
48     age_group = classify_age(age)
49     print(f"You are classified as: {age_group}")
50 # Example usage:
51 if __name__ == "__main__":
52     age = int(input("Enter your age: "))
53     age_group = classify_age(age)
54     print(f"You are classified as: {age_group}")
55 # Example usage:
56 if __name__ == "__main__":
57     age = int(input("Enter your age: "))
58     age_group = classify_age(age)
59     print(f"You are classified as: {age_group}")
60 # Example usage:
61 if __name__ == "__main__":
62     age = int(input("Enter your age: "))
63     age_group = classify_age(age)
64     print(f"You are classified as: {age_group}")
65

```

The terminal output shows the execution of the script with the following input and output:

```

PS C:\Users\Sai Pallavi\Desktop\AI(LAB)> & "c:\Users\Sai Pallavi\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Sai Pallavi/Desktop/AI(LAB)/age_classification.py"
Enter your age: 5
You are classified as: Child
Enter your age: 15
You are classified as: Teenager
Enter your age: 30
You are classified as: Adult
Enter your age: 70
You are classified as: Senior

```

Analysis of Conditional Logic (if- Elif -else):

- The conditions are checked one by one from top to bottom, and only the first matching block is executed.
- Invalid ages, such as negative values, are handled at the beginning to avoid incorrect classification.
- Each elif condition clearly defines an age range, ensuring there is no overlap between categories.
- The structure is simple and easy to understand, especially for beginners.
- This approach works well when the classification rules are straightforward and follow a clear sequence.

Explanation of the Alternative Logic:

- Age ranges are stored as range objects mapped to their respective age groups in a dictionary.
- The function iterates through the dictionary and checks which range the age belongs to.
- This separates data (age ranges) from logic, making the code more modular.
- Easy to modify or extend by just updating the dictionary.
- Useful when classification rules may change or grow over time.

Task Description #4: For and While Loops (Sum of First n Numbers)

Scenario: You need to calculate the sum of the first n natural numbers.

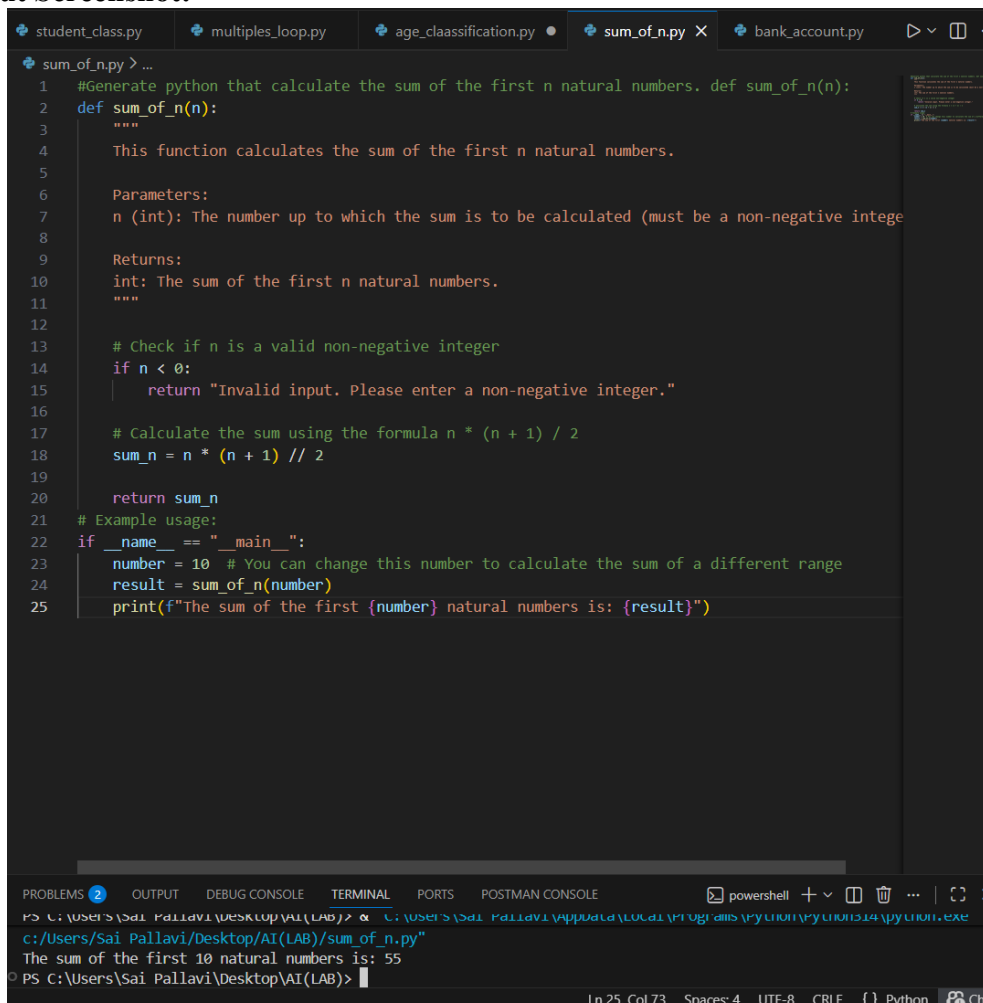
Task:

- Use AI assistance to generate a `sum_to_n()` function using a for loop.
- Analyze the generated code.
- Ask the AI to suggest an alternative implementation using a while loop or a mathematical formula.

Expected Output #4

- Python function to compute the sum of first n numbers.
- Correct output for sample inputs.
- Explanation and comparison of different approaches

Output Screenshot:



```
student_class.py  multiples_loop.py  age_classification.py  sum_of_n.py X  bank_account.py  ▸ ▾ □ ▢ ...
sum_of_n.py > ...
1  #Generate python that calculate the sum of the first n natural numbers. def sum_of_n(n):
2  def sum_of_n(n):
3      """
4      This function calculates the sum of the first n natural numbers.
5
6      Parameters:
7      n (int): The number up to which the sum is to be calculated (must be a non-negative integer)
8
9      Returns:
10     int: The sum of the first n natural numbers.
11     """
12
13     # Check if n is a valid non-negative integer
14     if n < 0:
15         return "Invalid input. Please enter a non-negative integer."
16
17     # Calculate the sum using the formula n * (n + 1) / 2
18     sum_n = n * (n + 1) // 2
19
20     return sum_n
21
22 # Example usage:
23 if __name__ == "__main__":
24     number = 10 # You can change this number to calculate the sum of a different range
25     result = sum_of_n(number)
26     print(f"The sum of the first {number} natural numbers is: {result}")
```

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```
PS C:\Users\Sai Pallavi\Desktop\AI(LAB)> cd C:\Users\Sai Pallavi\AppData\Local\Programs\Python\Python134\python.exe
C:\Users\Sai Pallavi\Desktop\AI(LAB)\sum_of_n.py
The sum of the first 10 natural numbers is: 55
PS C:\Users\Sai Pallavi\Desktop\AI(LAB)>
```

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Analysis of the for Loop Code:

- The loop runs from 1 to n, adding each number to the variable total.
- Initialization of total to 0 ensures correct accumulation.
- The logic is simple, clear, and easy to understand.
- This approach works well when the number of iterations is known.

Task Description #5: Classes (Bank Account Class)

Scenario: You are designing a basic banking application.

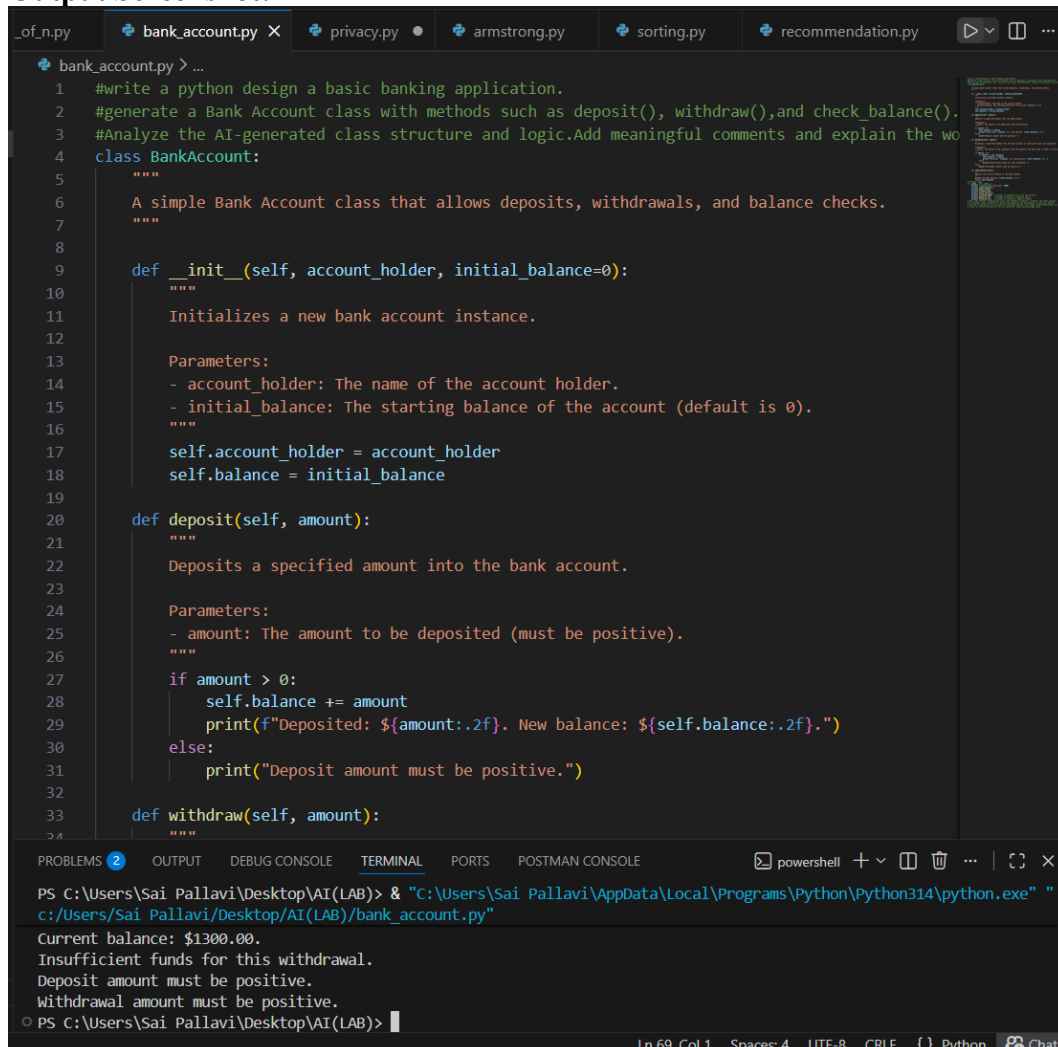
Task:

- Use AI tools to generate a Bank Account class with methods such as deposit(), withdraw(), and check balance().
- Analyze the AI-generated class structure and logic.
- Add meaningful comments and explain the working of the code.

Expected Output #5

- Complete Python Bank Account class.
- Demonstration of deposit and withdrawal operations with updated balance.
- Well-commented code with a clear explanation

Output Screenshot:



```
bank_account.py X privacy.py armstrong.py sorting.py recommendation.py
bank_account.py > ...
1 #write a python design a basic banking application.
2 #generate a Bank Account class with methods such as deposit(), withdraw(),and check balance().
3 #Analyze the AI-generated class structure and logic.Add meaningful comments and explain the wo
4 class BankAccount:
5     """
6     A simple Bank Account class that allows deposits, withdrawals, and balance checks.
7     """
8
9     def __init__(self, account_holder, initial_balance=0):
10        """
11        Initializes a new bank account instance.
12
13        Parameters:
14        - account_holder: The name of the account holder.
15        - initial_balance: The starting balance of the account (default is 0).
16        """
17        self.account_holder = account_holder
18        self.balance = initial_balance
19
20    def deposit(self, amount):
21        """
22        Deposits a specified amount into the bank account.
23
24        Parameters:
25        - amount: The amount to be deposited (must be positive).
26        """
27        if amount > 0:
28            self.balance += amount
29            print(f"Deposited: ${amount:.2f}. New balance: ${self.balance:.2f}.")
30        else:
31            print("Deposit amount must be positive.")
32
33    def withdraw(self, amount):
34        """
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Analysis of Class Structure and Logic:

- **The Bank Account class represents a real-world bank account.**
- **The constructor (`__init__`) initializes account details and balance.**
- **The `deposit()` method ensures only positive amounts are added.**
- **The `withdraw()` method checks for valid amount and sufficient balance.**
- **The `check_balance()` method displays the current balance.**
- **Input validation prevents invalid transactions.**

Explanation of Code Working:

- **A Bank Account object is created with an initial balance to represent a real bank account.**
- **When a deposit is made, the amount is first validated and then added to the existing balance.**
- **During withdrawal, the code checks whether enough balance is available before deducting the amount.**
- **After each transaction, the updated balance is displayed to show the current account status.**
- **Overall, the code is easy to read, well-structured, and follows good programming practices.**